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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR

EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course – MG1DSCCFS100 – INTRODUCTION TO DIGITAL FORENSICS AND TOOLS

(2024 ADMISSION ONWARDS)

SCHEME

Duration: 1.5 Hours Maximum Marks: 50

Part A

Short Answer Type Questions

Answer any 10 questions

Each question carries 1 mark

1. Describe File carving.
   Extracting data from unallocated space on a hard drive is called File carving. It can be done manually or with the help of a tool.

2. Explain the term forensic science.
   Forensic science is the application of scientific methods or expertise to investigate crimes or examine evidence that might be presented in a court of law.

3. Describe link files.
   Link files are simply shortcuts to files. They point to other files. Link files can be created by user, or by the computer.

4. Describe Metadata.
   Metadata means data about data. File system metadata include the date and time a file or folder was created, accessed, or modified.

5. Explain Removable storage media.
   Removable media is **any type of storage device that can be removed from a computer** while the system is running. Removable storage media include things like DVDs, external hard drives, thumb drives, and memory cards.

6. Summarize the role of forensic examiners in the judicial system.
   The digital forensics practitioner most often plays the role of an expert witness. An expert is someone who can assist the judge or jury to understand and interpret evidence they may be unfamiliar with.

7. Interpret the concept of eDiscovery.
   eDiscovery is defined as “The process of collecting, preparing, reviewing, and producing electronically stored information (“ESI”) in the context of the legal process”.

8. Outline the concept of obfuscation.
   Obfuscation refers to the process of concealing something important, valuable or critical. Obfuscation is intended to make things harder to understand. It is used to protect computer code, rather than the data itself. Obfuscation also protects code from reverse engineering.

9. Outline the Fourth amendment in digital forensics is important? Illustrate the significance.
   The Fourth Amendment says: “The right of the people to be secure in their persons, houses, papers, and effects, against unreasonable searches and seizures, shall not be violated, and no Warrants shall issue, but upon probable cause, supported by Oath or affirmation, and particularly describing the place to be searched, and the persons or things to be seized.”
   Any evidence deemed to be seized in violation of the Fourth Amendment is inadmissible in a court of law.

10. Implement the need for using the FTK tool in digital forensics.
    FTK tool is used to create forensic images of evidence disk/drives. It also provides integrity checking function which generates a hash report and matches the hash of the evidence before and after creating the image of the original evidence.

11. Categorize the forensic tools used to collect data.
    Autopsy, FTK Imager etc are forensic tools used to collect data.

12. What is FDAC?
    FDAC is a data carving tool. It carves data from unallocated space and slack space.

Part B

Short Essay Type Questions

Answer any 4 questions

Each question carries 5 marks

13. Explain in detail the Windows registry structure.

    Windows registry is a database for configuration files. The registry keeps track of user and system configuration and preferences.

    The five main Hives in Windows Registry are **HKEY_CLASSES_ROOT, HKEY_CURRENT_USER, HKEY_LOCAL_MACHINE, HKEY_USERS** and **HKEY_CURRENT_CONFIG.**

    Explain 5 hives – **1 mark for each**. Total **5 marks**.

14. Explain policies and procedures involved in handling evidence.

    The tasks involving how the labs handles evidence, conducts examinations, keeps records, and secures its facility should be governed by policies and Standard Operating Procedures (SOPs).

    SOPs are documents that detail, among other things, how common forensic examinations should be performed.

    Explain – **5 marks**.

15. Discuss the two options to collect evidence from a cell phone.

    - **Turn the phone off.** The concern with this approach is that, the phone may be password-protected. If the phone is turned off, the investigators need find out the password to access the phone at the laboratory. This is time consuming. [**2.5 marks**]
    - **Place the phone in special containers that shield the phone from wireless signals.** Empty paint cans and Faraday bags are two of the more typical choices. Both of these items are effective at safeguarding the phone from cell signals. If possible, it may be best to isolate the phone in a Faraday bag or arson can and leave it powered on. It can then be transported to the lab to be examined in a shielded room. [**2.5 marks**]

16. Explain Hashing.

    Define hashing – **1 mark**.

    Mention types of hashing algorithms – **2 marks**.

    Uses of hashing – **2 marks**.

17. Explain the concept of FDE.
    FDE stands for Full Disk Encryption. Encrypting an entire drive is called Full Disk Encryption. Explain – **2.5 marks**.

    Drawbacks of FDE – **2.5 marks**.

18. What is PD Server?
    PD server refers to a **"Perception and Data Server"** or **"Processing and Distribution Server"** used for managing, processing, and analysing digital evidence in a forensics investigation.
    These servers are part of a larger digital forensics infrastructure and provide functionalities such as Data Collection, Data processing, Data storage, Analysis and Reporting, Collaboration and distribution.
    Explain – **5 marks**.

Part C

Essay Type Questions
Answer any 2 questions
Each question carries 10 marks

19. Explain in detail about Hibernation file.
    Hibernation – A power-off mode or ”cybernap” in which the computer saves its last state. - **1 mark**
    Three modes of hibernation – Sleep, Hibernation, Hybrid sleep – Explain in detail – **3 marks for each mode = 9 marks**.

20. Describe about evidence storage.
    Evidence storage in digital forensics refers to the secure and controlled process of storing digital evidence so that it remains intact and unaltered during the course of an investigation.
    Proper evidence storage is critical to ensure the integrity and admissibility of the evidence in legal proceedings.
        - Definition & description – **2 marks**.
    Key concepts involved in evidence storage include ;
    - Chain of custody – Labelling, documentation, tamper-proof seals etc. [**2 marks**]
    - Physical storage – Lockable containers, Faraday bags, restricted access etc. [**2 marks**]
    - Digital evidence storage – Forensic copies, write-protection, hashing. [**2 marks**]
    - Digital evidence repositories – Forensic servers like PD server, cloud storage etc. [**2 marks**]

21. Explain one of the network investigative tool and illustrate different types of networks.
    Tool explanation – any one tool (Wireshark, NetIntercept, Netwitness Investigator, Snort). [**4 marks**]

    Types of networks – LAN, WAN, MAN, PAN, CAN, GAN [**1 mark each – 6 marks**]

END OF SCHEME